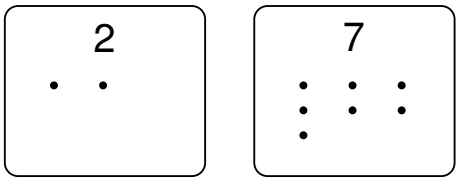
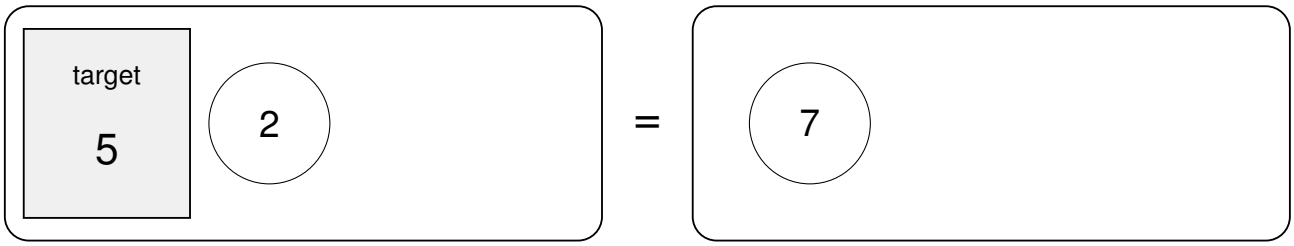
Keep the target on the left. Each weight may go on either pan or stay off. Use each weight at most once. Equal totals balance.

Target 5; weights 2 and 7



put 2 beside the target



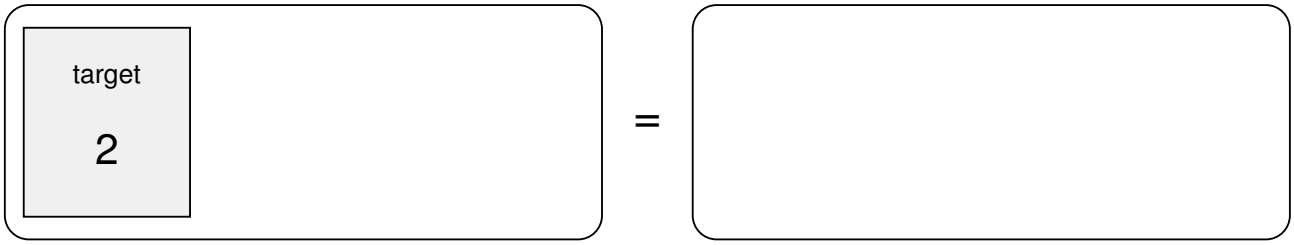
5 + 2 = 7

Problem 1: Compare the kits 1 and 2, 1 and 3, and 1 and 4. Which targets from 1 through 6 can each kit balance?

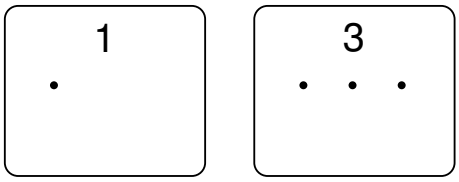
1 and 2 | 1 | 2 | 3 | 4 | 5 | 6 |

1 and 3 | 1 | 2 | 3 | 4 | 5 | 6 |

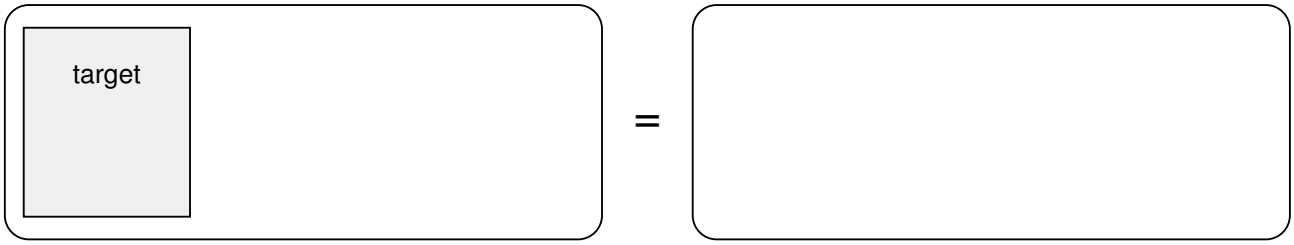
1 and 4 | 1 | 2 | 3 | 4 | 5 | 6 |

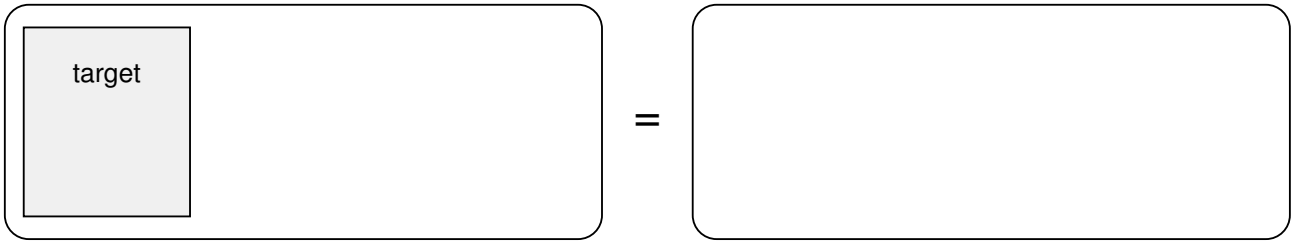


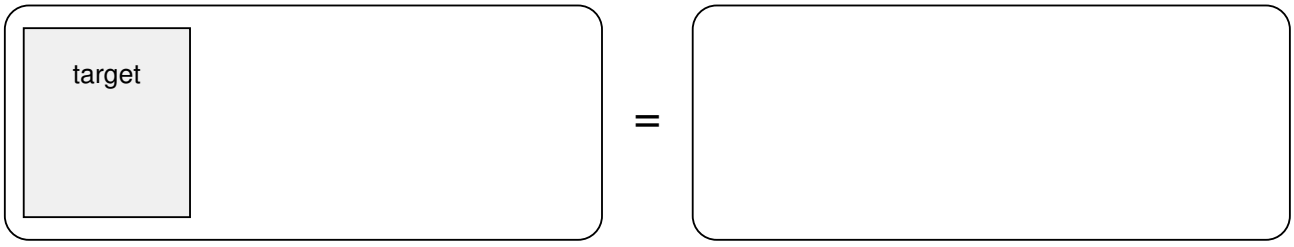
Problem 2: You have weights 1 and 3. Choose one more whole-number weight so that the kit balances every target from 1 through 13.



1	2	3	4	5	6	7
8	9	10	11	12	13	







Problem 3: Compare these kits: 1, 3, 8; 1, 3, 9; 1, 3, 10. For each, find the first target it cannot balance.

1, 3, 8

1	2	3	4	5	6	7
8	9	10	11	12	13	14

1, 3, 9

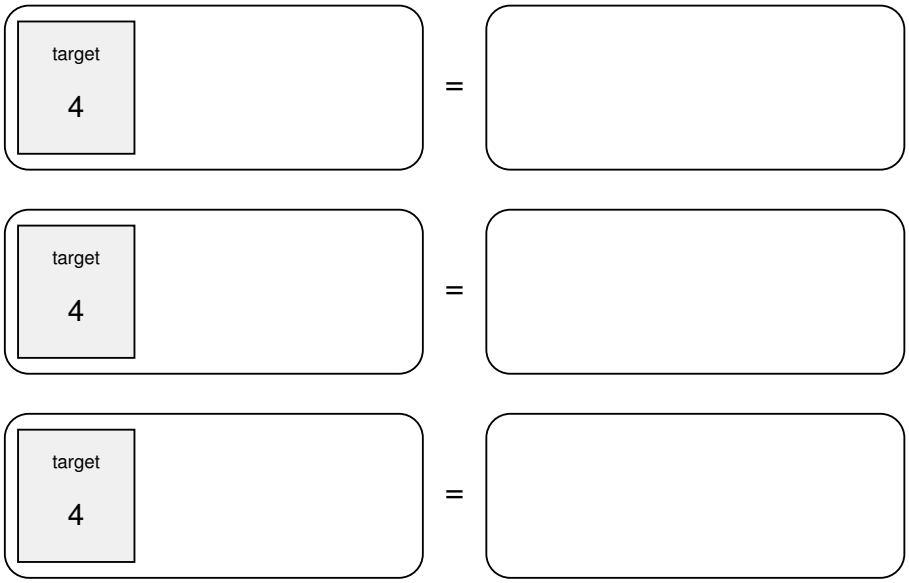
1	2	3	4	5	6	7
8	9	10	11	12	13	14

1, 3, 10

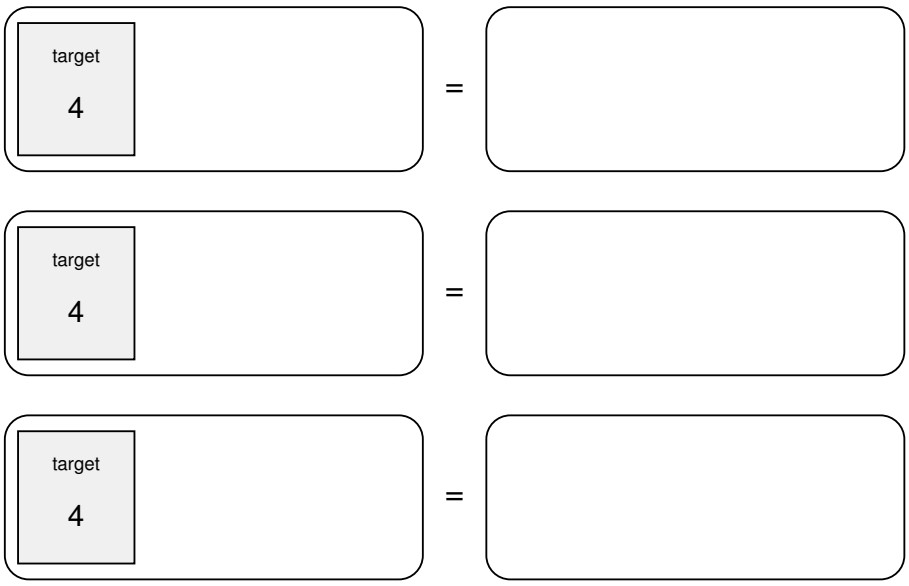
1	2	3	4	5	6	7
8	9	10	11	12	13	14

Problem 4: Use weights 1, 3 and 8 to balance 4 in every possible way. Then find every way to balance 4 with 1, 3 and 9.

1, 3, 8



1, 3, 9



Problem 5: Could a different three-weight kit balance every target from 1 through 14? Find one or explain why there cannot be one.

Problem 6: Choose a fourth weight to add to 1, 3 and 9. Make the longest possible unbroken run of targets starting at 1, and explain why your kit reaches all of them.